Structured analysis

0. Task title

"Investigation and Mitigation of UART Communication Interference between the Nicla Vision and a SLAM Robot"

1. Context diagram

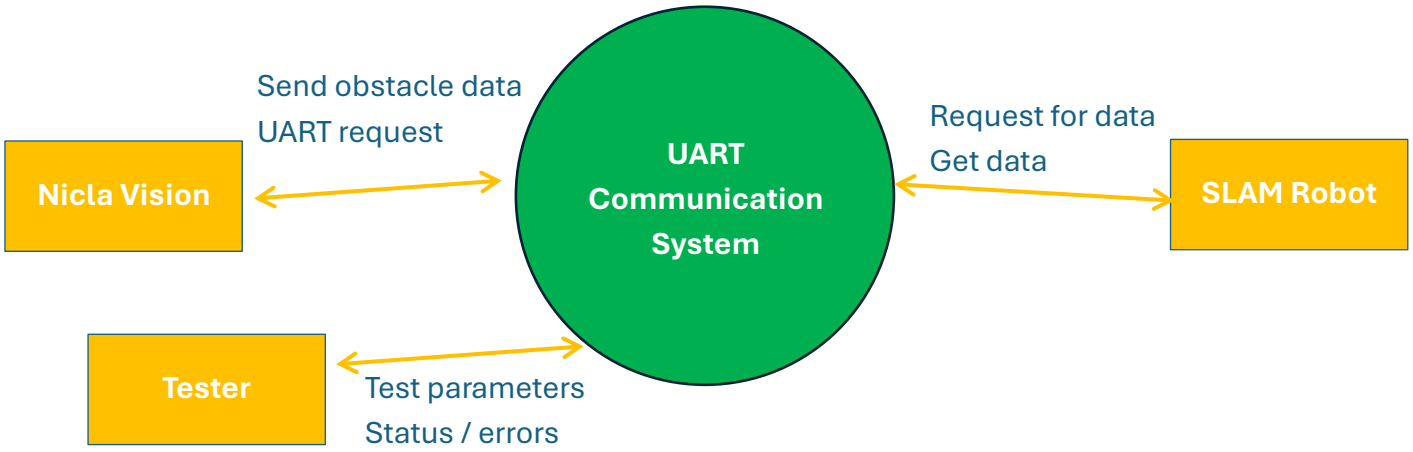
1a. Functional description of the complete system

The purpose of the system is to provide reliable communication of obstacle detection data from the Nicla Vision camera to the SLAM robot. The system will receive data transmitted by the Nicla Vision, determine whether the communication is valid, and send the valid obstacle information to the robot. The system will also identify and report communication errors and provide information that can be used to evaluate communication reliability under different operating conditions. When different operating conditions have been tested, a closer look at the biggest cause of errors will be investigated, and thus an attempt at removing the error.

1b. System information I/O

Terminator	Information to system	Information from system
Nicla Vision	Obstacle detection data Distance, relative position, width UART sequence number	UART request
SLAM Robot	Request for obstacle data	Valid obstacle data
Tester	Configuration or test parameters	Communication errors Timeout status Missing packets

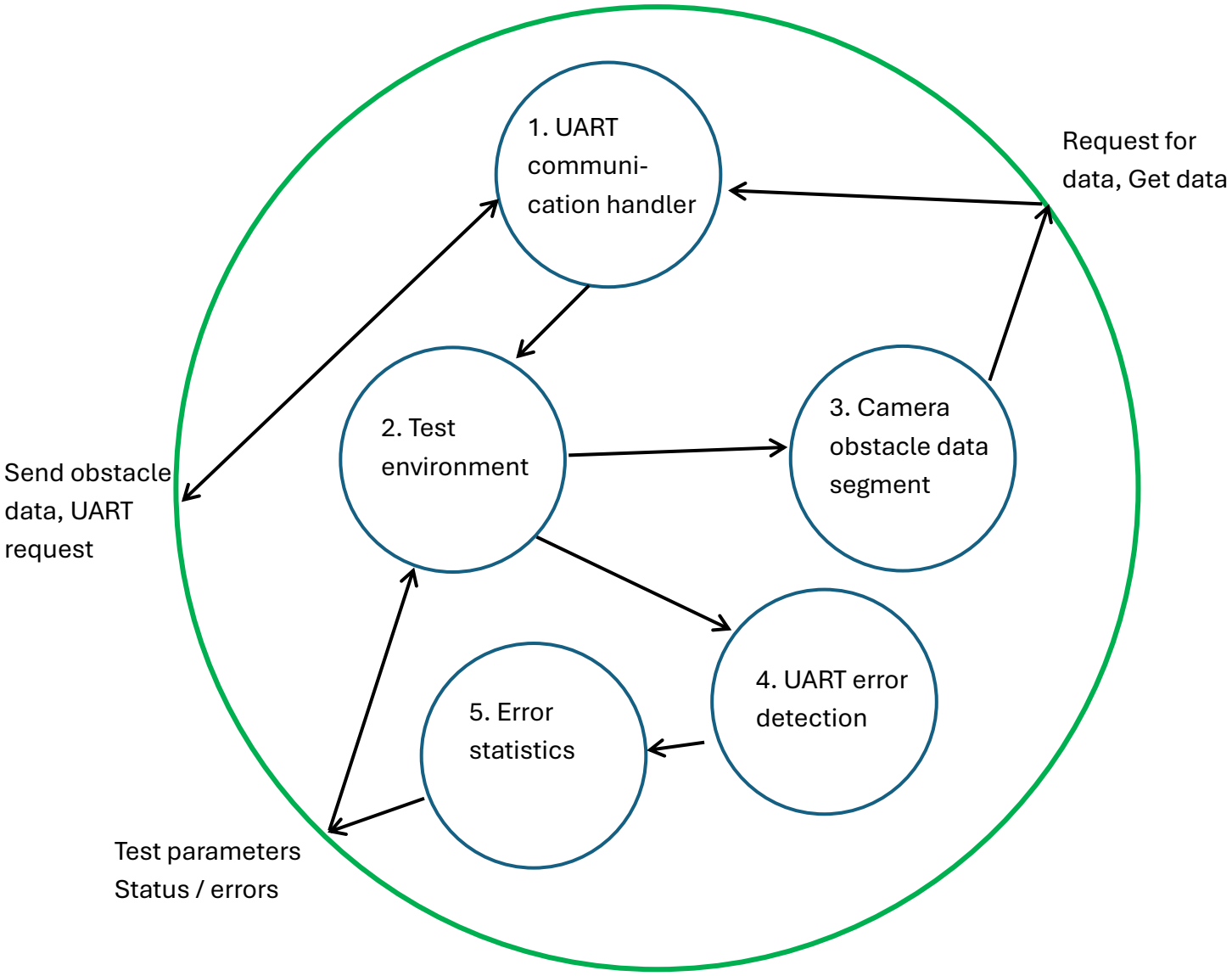
Context diagram



Event list

Terminators	Event
Nicla Vision	Receives request for obstacle packet, Sends obstacle packet and sequence number.
SLAM Robot	Sends request for obstacle detection data packet, Uses data to navigate
Tester	Set test parameters, Receives statistics of communication errors.

2. Inner analysis



2.1 UART communication handler

Handles communication between camera and robot (initializing). Sends request from robot to Nicla Vision, and the obstacle data from the camera to processing.

Input	Output
UART request from Robot UART data packet from Nicla Vision.	Processed UART request from Robot Obstacle data from Nicla Vision

2.2 Test environment

A test environment for more consistent tests. Uses test parameters to test factors that can affect the UART communication. A test will exist of 1000 packets sent from the camera.

Input	Output
Obstacle data packets, with a sequence number. Test parameters	Valid obstacle data. Invalid packets Time segments Missing sequence numbers

2.3 Camera obstacle data segment

Processes valid obstacle detection data and makes the information available to the SLAM robot for further use.

Input	Output
Valid obstacle data.	A segment, consisting of estimation f x position, width and depth.

2.4 UART error detection

Identifies missing, corrupted, incomplete, or invalid communication between the Nicla Vision and the robot.

Input	Output
Invalid packets / received data packets Time segments Missing sequence numbers	Type of errors Timeouts Missing packets

2.5 Error statistics

Collects information about communication performance in a table, so that reliability can be evaluated under different operating conditions.

Input	Output
Type of errors Timeouts Missing packets	Packet failure rate Error statistics Test results.